Incremental Graph Code (IncrCode)

FILE EDIT PROJECT INFO

File Explorer

task_tracker.igc

Graph Editor

ADD NODE

Code Editor

```
{
    "nodes": [
        {
            "id": "start",
            "data": {
                "label": "Start"
            },
            "width": 20,
            "height": 20
        }
    ],
    "edges": []
}
```

Session Configuration

START NEW SESSION

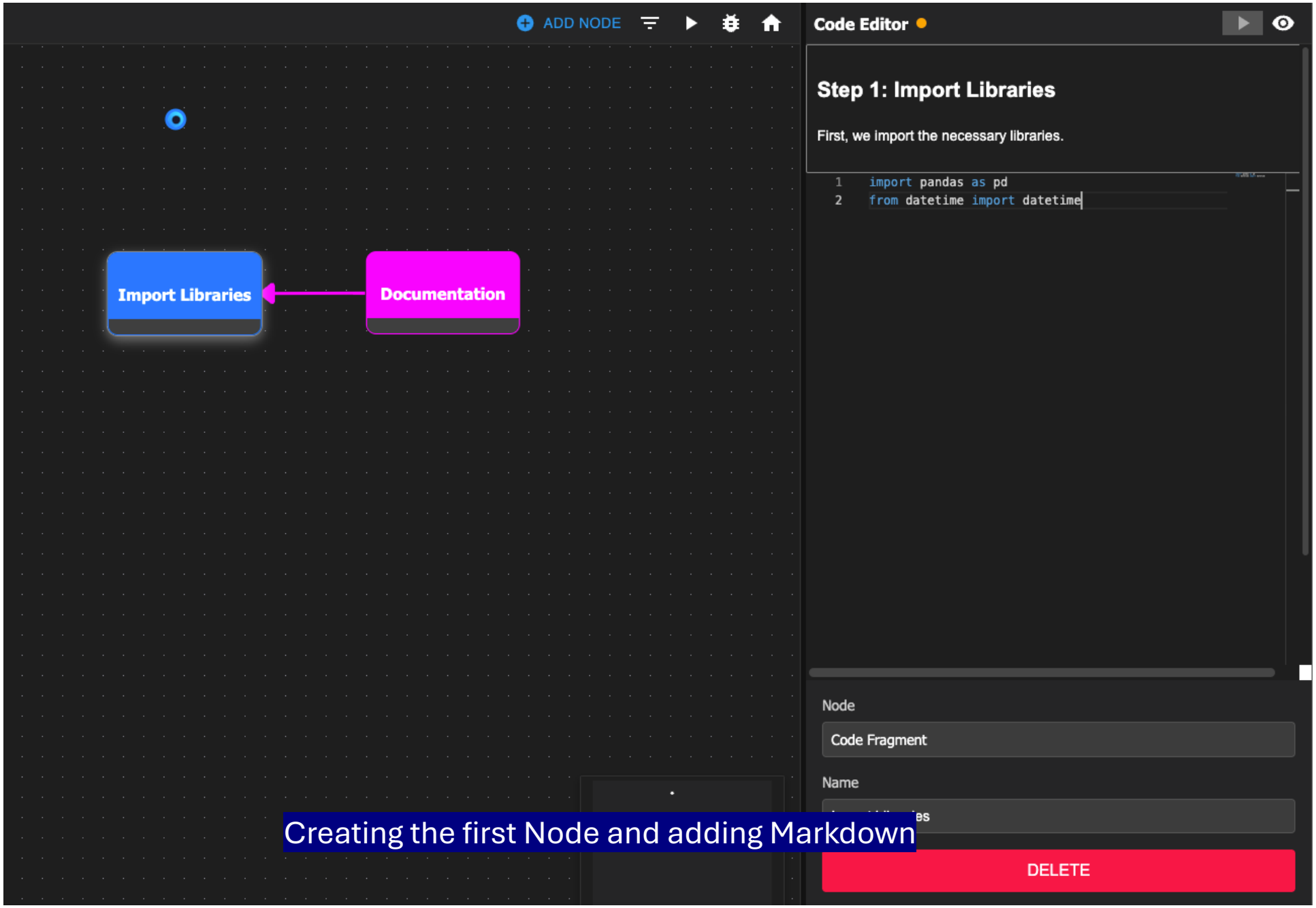
Select Session

Select

No Configurations

Initial Screen when creating a new project in IGC

Incremental Graph Code



Creating the first Node and adding Markdown

Incremental Graph Code (IncrCode)

FILE EDIT PROJECT INFO

File Explorer

task_tracker.igc

Graph Editor

ADD NODE

Import Libraries

Documentation

Input Data

Calculate Total Time Spent

Calculate Time Spent per Day

Most Time-Consuming Task

Suggestions

Code Editor

Step 1: Import Libraries

First, we import the necessary libraries.

```
import pandas as pd
from datetime import datetime
```

Node

Code Fragment

Name

Import Libraries

DELETE

Session Configuration

START NEW SESSION

Select Session

Select

No Configurations

Created all Nodes for the execution proposed task

Incremental Graph Code

Incremental Graph Code (IncrCode)

FILE EDIT PROJECT INFO

File Explorer

.sessions

task_tracker.igc

Graph Editor

ADD NODE

Import Libraries

Documentation

Input Data

Calculate Total Time Spent

Calculate Time Spent per Day

Most Time-Consuming Task

Suggestions

Code Editor

Step 1: Import Libraries

First, we import the necessary libraries.

```
import pandas as pd
from datetime import datetime
```

OUTPUT ERRORS CONFIGURATION METRICS

```
{
  "pd": "<module>",
  "datetime": "<class>"
}
```

Node

Code Fragment

Name

Import Libraries

DELETE

Session Configuration

START NEW SESSION

Select Session

task_tracker_execution1

```
{
  "pd": "<module>",
  "datetime": "<class>"
}
```

We create a new session named “task_tracker_execution1”. Then we execute the first Node. You can see the session configuration has also changed.

Session Configuration

START NEW SESSION

Select Session

task_tracker_execution1

```
{
  "pd": "<module>",
  "datetime": "<class>"
}
```

Here specifically is the data of the session configuration

Incremental Graph Code (IncrCode)

FILE EDIT PROJECT INFO

File Explorer

task_tracker.igc

Session Configuration

START NEW SESSION

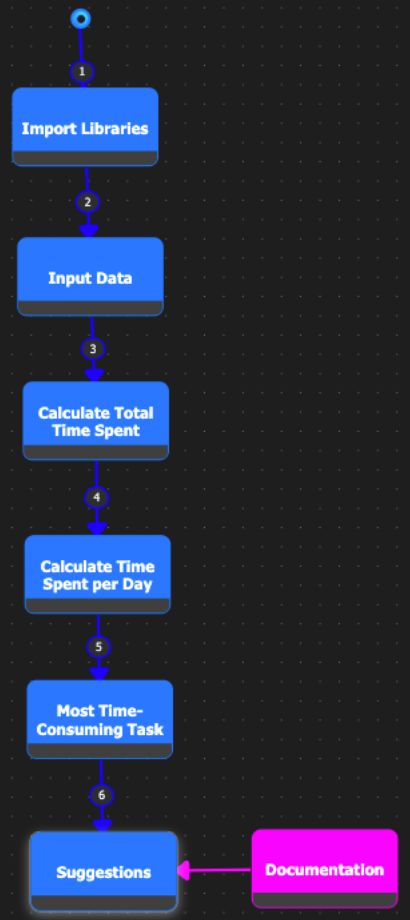
Select Session

task_tracker_execution1


```
{
  "pd": "<module>",
  "datetime": "<class>",
  "data": {
    "Task": [
      "Email follow-up",
      "Project meeting",
      "Code review",
      "Write report",
      "Team lunch",
      "Client call"
    ],
    "Date": [
      "2024-07-01",
      "2024-07-01",
      "2024-07-02",
      "2024-07-02",
      "2024-07-03",
      "2024-07-03"
    ],
    "Duration (hours)": [
      1,
      2,
      1.5,
      3,
      1,
      1
    ]
  },
  "tasks": "<DataFrame>",
  "total_time_spent": 9.5,
  "time_per_day": "<Series>",
  "most_time_consuming_task": "<
  "time_saving_suggestion": 0.30
}
```


Graph Editor

ADD NODE



Code Editor

Step 6: Suggestions for Task Management

Finally, we provide a simple suggestion to save time by reducing the duratic of the most time-consuming task.

```
# Simple suggestion to improve task management b
time_saving_suggestion = most_time_consuming_tas
print(f"Suggestion: Save {time_saving_suggestior
```

OUTPUT ERRORS CONFIGURATION METRICS

```
Suggestion: Save 0.3 hours by reducing the time sp
ent on 'Write report' by 10%.
```

Node

Code Fragment

Name

Suggestions

DELETE

We execute all cells here. Notice the session configuration contains much more data. We see the STDOUT in the terminal under the code editor.

File Explorer

.sessions

Session Configuration

START NEW SESSION

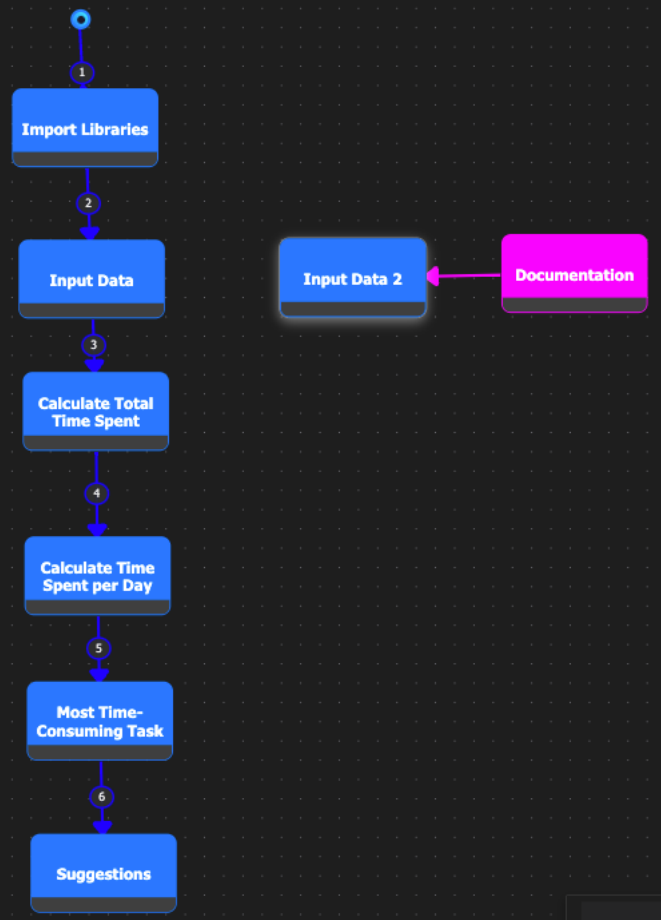
Select Session

task_tracker_execution1


```
{
  "pd": "<module>",
  "datetime": "<class>",
  "data": {
    "Task": [
      "Email follow-up",
      "Project meeting",
      "Code review",
      "Write report",
      "Team lunch",
      "Client call"
    ],
    "Date": [
      "2024-07-01",
      "2024-07-01",
      "2024-07-02",
      "2024-07-02",
      "2024-07-03",
      "2024-07-03"
    ],
    "Duration (hours)": [
      1,
      2,
      1.5,
      3,
      1,
      1
    ]
  },
  "tasks": "<DataFrame>",
  "total_time_spent": 9.5,
  "time_per_day": "<Series>",
  "most_time_consuming_task": "<
  "time_saving_suggestion": 0.30
}
```


Graph Editor

ADD NODE



Code Editor

Step 2: Input New Data

(Different Data) This is a new version of the input data we want to run.

```
# New sample data for daily tasks
data = {
    'Task': ['Morning Jog', 'Team Meeting', 'Coc
    'Date': ['2024-07-01', '2024-07-01', '2024-(
    'Duration (hours)': [1, 2, 2.5, 1.5, 3, 0.5]
}

# Create a DataFrame
tasks = pd.DataFrame(data)
tasks['Date'] = pd.to_datetime(tasks['Date'])
print(tasks)
```

OUTPUT ERRORS CONFIGURATION METRICS

```
    2.5
       3     Lunch with Client 2024-07-02
     1.5
        4  Prepare Presentation 2024-07-03
      3.0
         5      Client Follow-up 2024-07-03
       0.5
```

Node

Code Fragment

Name

Input Data 2

DELETE

We now create a new node with new input data.

Incremental Graph Code

Incremental Graph Code (IncrCode)

FILE EDIT PROJECT INFO

File Explorer

.sessions

Session Configuration

START NEW SESSION

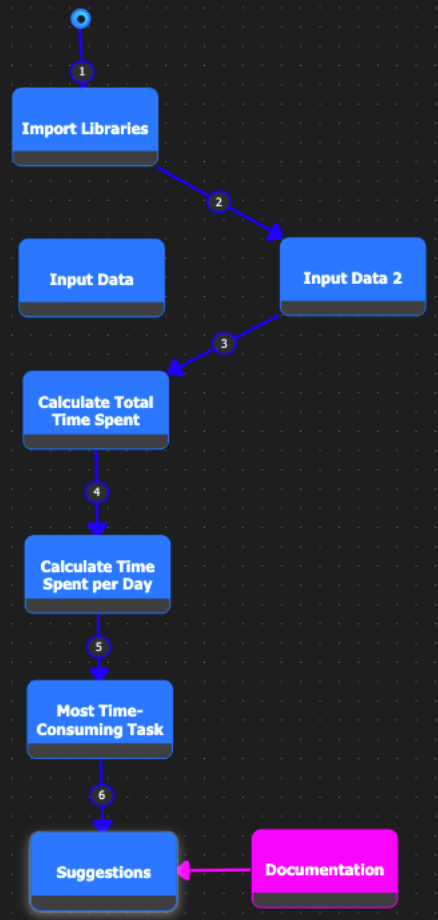
Select Session

task_tracker_execution2


```
{
  "pd": "<module>",
  "datetime": "<class>",
  "data": {
    "Task": [
      "Morning Jog",
      "Team Meeting",
      "Code Implementation",
      "Lunch with Client",
      "Prepare Presentation",
      "Client Follow-up"
    ],
    "Date": [
      "2024-07-01",
      "2024-07-01",
      "2024-07-02",
      "2024-07-02",
      "2024-07-03",
      "2024-07-03"
    ],
    "Duration (hours)": [
      1,
      2,
      2.5,
      1.5,
      3,
      0.5
    ]
  },
  "tasks": "<DataFrame>",
  "total_time_spent": 10.5,
  "time_per_day": "<Series>",
  "most_time_consuming_task": "<
  "time_saving_suggestion": 0.30
}
```


Graph Editor

ADD NODE



Code Editor

Step 6: Suggestions for Task Management

Finally, we provide a simple suggestion to save time by reducing the duratic of the most time-consuming task.

```
# Simple suggestion to improve task management b
time_saving_suggestion = most_time_consuming_tas
print(f"Suggestion: Save {time_saving_suggestior
```

OUTPUT ERRORS CONFIGURATION METRICS

```
Suggestion: Save 0.3 hours by reducing the time sp
ent on 'Prepare Presentation' by 10%.
```

Node

Code Fragment

Name

Suggestions

DELETE

We create a new session (named “task_tracker_execution2”) and we create an execution path through “Input Data 2” now.

Incremental Graph Code (IncrCode)

FILE EDIT PROJECT INFO

File Explorer

.sessions

Session Configuration

START NEW SESSION

Select Session

task_tracker_execution2

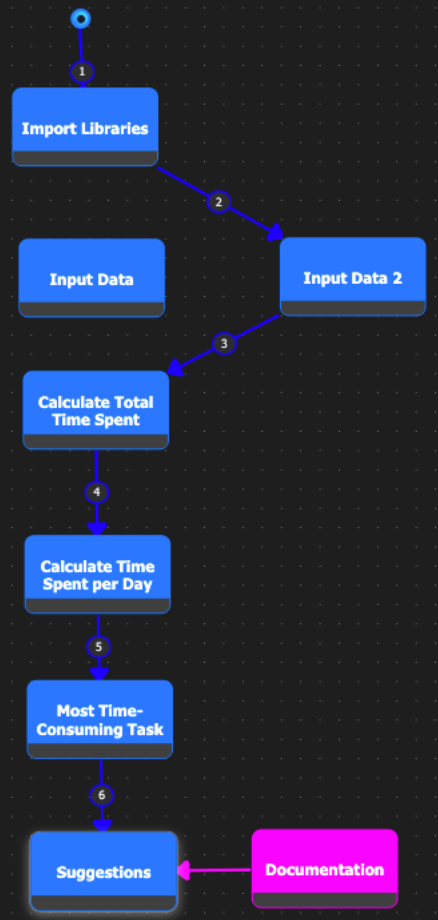
task_tracker_execution1

task_tracker_execution2


```
  "pd": "<module>",
  "datetime": "<class>",
  "data": {
    "Task": [
      "Morning Jog",
      "Team Meeting",
      "Code Implementation",
      "Lunch with Client",
      "Prepare Presentation",
      "Client Follow-up"
    ],
    "Date": [
      "2024-07-01",
      "2024-07-01",
      "2024-07-02",
      "2024-07-02",
      "2024-07-03",
      "2024-07-03"
    ],
    "Duration (hours)": [
      1,
      2,
      2.5,
      1.5,
      3,
      0.5
    ]
  },
  "tasks": "<DataFrame>",
  "total_time_spent": 10.5,
  "time_per_day": "<Series>",
  "most_time_consuming_task": "<
  "time_saving_suggestion": 0.30
}
```


Graph Editor

ADD NODE



Code Editor

Step 6: Suggestions for Task Management

Finally, we provide a simple suggestion to save time by reducing the duratic of the most time-consuming task.

```
# Simple suggestion to improve task management b
time_saving_suggestion = most_time_consuming_tas
print(f"Suggestion: Save {time_saving_suggestior
```

OUTPUT ERRORS CONFIGURATION METRICS

```
Suggestion: Save 0.3 hours by reducing the time sp
ent on 'Prepare Presentation' by 10%.
```

Node

Code Fragment

Name

Suggestions

DELETE

We now can switch back and forth between the two different sessions to compare outputs and state.

Incremental Graph Code